

0.1 Section 1.7 Intro to Proofs

What do we prove? Usually that some pattern or formula is true.

Example 1

$$\begin{aligned}6 + 8 &= 14 \\12 + 16 &= 28 \\2 + 2 &= 4\end{aligned}$$

We believe that the sum of 2 even integers is even.

Try to prove that: even + even = even

Work An even number can be defined as $2n$, where n is an integer

$$2n + 2m = 2(n + m)$$

Proof. let a, b be even integers.

Then there exists integers n, m such that $a = 2n, b = 2m$

$$a + b = 2n + 2m = 2(m + n)$$

is even, as desired

□

But writing it this way is too long. Let's define some notation:

0.1.1 Set Notation

Name	Symbol	Set Notation
Set of all natural numbers	$\mathbb{N}$	$\{0, 1, 2, 3, \dots\}$
Set of all integers	$\mathbb{Z}$	$\{0, -1, 1, -2, 2, \dots\}$
Set of positive integers	$\mathbb{Z}^+$	$\{1, 2, 3, \dots\}$
Set of negative integers	$\mathbb{Z}^-$	$\{-1, -2, -3, \dots\}$
Set of rational numbers	$\mathbb{Q}$	$\{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\}$
Set of real numbers	$\mathbb{R}$	Limit of all convergent sequences of real numbers
Set of all complex numbers	$\mathbb{C}$	$\{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$

$\exists$ means there exists

Let's rewrite the previous example using this new notation

$$a, b \in \mathbb{Z}, \text{ both even}$$

Then, $\exists m, n \in \mathbb{Z}$ such that

$$a = 2m, b = 2n$$

Their sum $a + b = 2m + 2n = 2(m + n)$
is even, as desired

0.1.2 Methods of Proving $p \implies q$

1. **Direct Proof:**

Start with p and *math* then conclude q

2. **Indirect Proof:**

Prove the contrapositive proof $\neg q \implies \neg p$

3. **Finite Cases:**

Use the fact that $p \equiv p_1 \vee p_2 \vee \dots \vee p_n$

Then prove $(p_1 \implies q) \wedge (p_2 \implies q) \wedge \dots \wedge (p_n \implies q)$